



McGill University

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Final Project: Optimizing Fresh Food Logistics for Nunavik

MGSC 662 – Decision Analytics

Group 1M

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1. Introduction

Above the 55th parallel in Quebec is Nunavik, a region with 14 remote Inuit communities. The reality for 60% of these people is facing food insecurity, meaning they lack access to a sufficient quantity of food, or lack food of adequate quality to meet their needs. Transporting food to these communities is a logistically difficult task. Roads and railways are unable to be used for 9 months out of the year due to the harsh weather conditions. Air transportation is the only reliable method of transportation year round, however, this is not perfect either. Limited airport infrastructure due to lack of demand means flights are infrequent and planes have limited cargo.

A central organization involved in the effort of transporting cargo to these communities is La Fédération des coopératives du Nouveau-Québec (FCNQ) - a federation of 14 Inuit-owned co-ops. FCNQ manages the sourcing, purchasing, and preparation of weekly food shipments for each community. All products first arrive at its central warehouse in Baie-D'Urfé near Montreal, where they are sorted, organized, and prepared for transport north to the FCNQ co-ops, which operate as the grocery stores in each community.

From the warehouse, the journey begins with refrigerated trucks, some owned by FCNQ and others operated by their partner Kepa Transport, travelling to either Montréal-Trudeau International Airport (YUL) or La Grande Rivière Airport (LGR). These long drives are often the only part of the supply chain where the food remains consistently refrigerated.

Once the trucks arrive at the airport, Air Inuit takes on the air transportation portion of the journey. As one of the main carriers serving Nunavik, it plays a crucial role in connecting the region's communities. A major challenge appears at this stage: the cargo holds on Air Inuit aircraft are not temperature-controlled. Fresh food is exposed to outside conditions for the entire flight, and delays on the tarmac or extended handling can cause temperature fluctuations that accelerate spoilage.

These difficulties are magnified for northern destinations such as Salluit. A shipment may pass through several hubs before arrival, accumulating idle time at each stop and increasing the likelihood of spoilage. Harsh winter weather also causes frequent cancellations and unplanned returns, and FCNQ estimates that 15 percent or more of certain fresh foods can arrive spoiled, with higher rates during winter months. This fragile cold chain makes weekly planning extremely challenging. FCNQ must decide how much of each product to ship while accounting for the fact

that some portion will not survive the journey. The goal for this report is to build a linear optimization model for FCNQ to find the optimal amount of food to send to each community. After accounting for food spoilage, every community will receive its demand for supplies for the week. The report will account for the plane's capacity, community demands, and distance between communities to help alleviate the food insecurity problem.

2. Problem Description and Formulation

2.1 Problem Description

To understand the FCNQ supply chain and identify where optimization is needed, multiple interviews and email exchanges were conducted with FCNQ staff. These discussions clarified the flow of goods through the network, the constraints FCNQ faces, and the parts of the system where the organization can meaningfully intervene.

The process begins when each of the 14 co-ops places its weekly food order. Demand is highly variable and depends on local events, seasonal activities, and community-specific conditions, meaning FCNQ must respond to needs that shift from week to week. Once an order is approved, FCNQ purchases the required products from major suppliers such as Loblaw's, Dole, Lactantia, or Chiquita. The suppliers ship the products to FCNQ's warehouse in Baie-D'Urfé near Montreal, where the items are sorted, packed, and labelled for each community. The warehouse, which was recently expanded, has ample capacity to receive, store, and prepare orders before they are transported north.

After the items are prepared, FCNQ must decide which airport to use for each shipment. Deliveries can be sent by refrigerated trucks operated by FCNQ or outsourced to Kepa Transport to one of two transshipment points. Montréal-Trudeau International Airport (YUL) is only a 20-minute drive from the warehouse and serves communities routed through the northern hubs of Kuujuaq or Puvirnituq. La Grande Rivière Airport (LGR), by contrast, requires a 13-hour truck drive of more than 1,300 km but is geographically closer to several northern communities and reduces total flight time. Choosing between YUL and LGR affects routing options, transit time, and expected spoilage.

Once food arrives at the airport, handling shifts to Air Inuit. At this stage, FCNQ no longer controls how or when the food is transported. Air Inuit determines whether a shipment travels directly to a community or first stops at a hub, and flight schedules are frequently affected by

weather, operational delays, and availability. A key limitation raised in every interview is that Air Inuit's cargo compartments are not temperature-controlled. Fresh food is exposed to outside conditions the entire time it is on the aircraft or waiting on the tarmac. Delays, reroutings, and multi-step flight paths all increase the amount of time food spends without refrigeration. According to federal sources, approximately 15 percent or more of fresh food arrives spoiled at some of these communities, with the rate rising significantly during winter.

Spoilage occurs at multiple stages of the network. During trucking, refrigeration may be interrupted when drivers take breaks. At the airport, food can sit for long periods before loading. In the air, the lack of temperature control leads to rapid degradation, and additional stops or weather-related disruptions prolong exposure. These issues are compounded by operational constraints: pallets take up fixed space, both weight and volume limits are strict for trucks and aircraft, and flight cancellations or returns are common during harsh weather.

This environment creates a fundamentally stochastic planning problem. FCNQ cannot control weather conditions, routing decisions, aircraft availability, or temperature exposure once shipments reach the airport. What FCNQ can control is the quantity of each product ordered, how much is held in the warehouse, how many trucks are dispatched, and how much of each item is sent to each community through each airport. Therefore, the model's objective is to minimize the total amount of food shipped while ensuring that the expected amount arriving at each community meets its weekly demand. This allows FCNQ to offset spoilage in a structured way and reduce over-ordering, ultimately supporting more reliable and affordable food access across Nunavik.

2.2 Parameters and Indices

To account for the unpredictability of flight schedules and transit times, the optimization model uses historical data to estimate expected travel times, spoilage rates, and cancellation probabilities. The key parameters include weekly demand for each community, the weight and spoilage rate of each food item, aircraft and truck capacities, available transportation arcs, and the probability of delays or rerouting. These differ by community and set of routes. The sets and parameters are defined as follows:

2.2.1 Sets and Parameters

C	Set of Communities (Destinations), indexed by c
H	Set of Entry Hubs (Montreal YUL ; La Grande LGR), indexed by h
R	Set of Valid Routes from Warehouse to Communities, indexed by r
P	Set of Products (Fruits & Vegetables), indexed by p , from NNC basket
$D_{c,p}$	Demand for product p in community c (lbs/week)
T_r	Estimated number of hours required to deliver goods along route r .
P(disruption)	Probability of a route disruption in a given week. Used to adjust expected deliveries.
σ_p	Spoilage rate of product p per hour (%)
ρ_p	Density of product p (lbs per pallet)
K_h^{Hub}	Maximum lift capacity at hub h (lbs/week)
K_W^{Truck}	Maximum weight capacity per truck (44, 092 lbs)
K_V^{Truck}	Maximum volume capacity per truck (26 pallets)

Given these Sets and Parameters, FCNQ has 2 key decision variables to take. How much FCNQ will send of each food item, for every community, to each airport, and how many trucks are needed to transport this food. Namely:

2.2.2 Decision Variables

x_r	Amount of product ordered at the warehouse for route r
N_h^{Truck}	Number of trucks required for entry hub h (Continuous variable for planning)

$m_{c,p}$	Slack variable representing missing demand (lbs) for product p in community c
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2.2.3 Main Model Formulation

$$\text{Minimize } Z = \sum_{r \in R} x_r + M \sum_{c \in C} \sum_{p \in P} m_{c,p}$$

Where M is a large penalty coefficient.

The objective is to deliver enough food to meet community needs with a protective buffer, while minimizing waste. Any shortfall is heavily penalized to reflect the critical importance of food access in the northern regions.

Subject to:

Constraint	Interpretation
$\sum_{r \in R c_r = c, p_r = p} (x_r \cdot S_r) + m_{c,p} \geq D_{c,p} \quad \forall c \in C, \forall p \in P$	This constraint makes sure that the usable food arriving, plus any unavoidable shortages, matches what the community needs
$\sum_{r \in R h_r = h} x_r \leq K_h^{\text{Hub}} \quad \forall h \in H$	The total weight of all food shipments assigned to a specific airport cannot exceed the total weekly flight capacity available at that hub
$\sum_{r \in R h_r = h} x_r \leq N^{\text{Truck}}_h \cdot K^{\text{Truck}}_W \quad \forall h \in H$	The total weight of food shipped to a hub cannot exceed the combined weight capacity of the trucks sent to that location

$\sum_{r \in R h_r = h} \frac{x_r}{\rho_{p_r}} \leq N^{\text{Truck}}_h \cdot K^{\text{Truck}}_V \quad \forall h \in H$	The total physical volume of the cargo, measured in pallet spots, cannot exceed the floor space available per truck
$\sum_{r \in R \text{origin} = W} x_r = \sum_{r \in R \text{dest} = c} x_r$ (adjusted for spoilage)	Food can't appear anywhere in the operation. It must originate from the warehouse
$x_r \geq 0 \quad \forall r \in R$	Non-negativity constraint for food items

3. Numerical Implementation and Results

3.1 Data Implementation

The optimization model was implemented in Python using Gurobi as the solver. The workflow follows a “physics-first” sequence: it begins with population estimates, translates these into consumption requirements, incorporates transit times across the network, and then applies spoilage dynamics for each product. Only after these physical components are established are they integrated into the final linear optimization model.

3.1.1 Demand Calculation

For each community-product pair (c, p) , weekly demand $D_{c,p}$ is computed based on:

- The 2024 population estimates for each of the 14 Nunavik communities (e.g. *Kuuujuaq* 2,800; *Puvirnituaq* 1,800; *Salluit* 1,500, etc.) (Ministère des Affaires municipales et de l'Habitation [MAMH], 2023)
- Per-capita annual consumption in kilograms for each product based out of the Nutrition North Canada (e.g. *0.895 kg / person for bananas*, *1.095 kg/person for apples*, etc.) (Government of Canada, 2024)

These are converted into weekly demand in pounds (lbs) using the conversion rate 1 kg = 2.20462 lb. Adding across all communities and products yields a total weekly fresh-produce demand of 107,392 lbs.

3.1.2 Network Topology

A hub and arc network is modelled with a single southern warehouse (YUL) and three northern “regions”:

- LGR Region (La Grande) serves the Southern Hudson Communities (Kuujuaraapik, Sanikiluaq, Umiujaq, Inukjuak)
- Puvirnituk Region serves the Northern Communities (Akulivik, Ivujivik, Salluit, plus Puvirnituk itself)
- Kuujuuaq Region serves the Ungava Bay Communities (Quaqtaq, Kangiqsualujjuaq, Tasiujaq, Aupaluk, Kangirsuk, Kangiqsujaq, plus Kuujuuaq itself)

A physical structure is introduced in the network by enforcing:

- Communities in the LGR Region must be supplied via the entry point LGR
- Communities in the Puvirnituk and Kuujuuaq regions are supplied via Montréal (YUL), which then connects to their respective hubs

This means that geography drives which entry point a community uses (*Appendix A*)

3.1.3 Risk and spoilage physics

For each directed arc (i, j) in the network the following is specified:

- A base transit time in hours combining physical distance, vehicle speeds, and Air Inuit flight schedules. (e.g. YUL → Kuujuuaq: 2.33 hours ; LGR → Sanikiluaq: 1.5 hours)
- A cancellation probability p_{ij} based on historical weather categorized as “extreme” in the Nunavik region using Meteostat API. Cancellation probability is proxied using the proportion of days with extreme weather conditions in the past year.

The effective travel time is computed by adjusting the nominal travel time upward to account for expected delays.

$$t_{ij}^{\text{eff}} = \frac{t_{ij}(1 + p_{ij})}{1 - p_{ij}} + \text{buffer}_{ij}$$

Where buffer_{ij} includes:

- Congestion-induced waiting time at YUL and LGR (4-6 hours on average), weighted by “fullness probabilities”
- A 45-minute handling buffer

Each product p has a spoilage rate per hour (e.g. bananas and grapes are fragile while turnips are tough). For every feasible route r (warehouse FCNQ → entry point → hub → community), a survival factor is pre-computed:

$$S_r = \prod_{\ell \in \text{legs}(r)} (1 - r_{p,\ell})^{t_\ell^{\text{eff}}}$$

where $r_{p,\ell}$ is the spoilage rate of product p on arc ℓ

To represent the refrigerated “freezer truck” into La Grande, the spoilage rate on the warehouse-to-LGR arc is reduced by 90% for all products. This captures the cold-chain advantage built into this leg of the model.

All survival rates are pre-computed in Python and passed into Gurobi as fixed coefficients, ensuring the optimization is linear.

3.1.4 Capacity Constraints

Two types of constraints are implemented:

1. Hub throughput constraints at YUL and LGR, limiting the total weight that can be handled per week
2. Dual truck capacity constraints (weight and volume) on the FCNQ warehouse to entry hubs
 - Weight capacity: Each long-haul truck is limited to 44,092 lbs of cargo (Transports Québec, 2023)
 - Volume capacity: Each trailer is limited to 26 pallet positions

The 26-pallet volume limit is derived from the geometry of the trailer and a standard CHEP pallet.

- A Standard CHEP pallet measure 48” x 40” (Performance Corp., 2023)
- A trailer internal floor measurement is 626” x 97” (Hale Trailer, 2022)

Using a straight load orientation, the number of pallet rows that fit along the length is computed:

Rows = Usable length / Pallet length = 626 / 48 ~ 13.04 → 13 rows

Total pallet capacity = 13 rows x 2 pallets wide = 26 pallets

3.1.5 Cancellation rates

Cancellation rates are estimated using the Meteostat API for each destination airport. The API allows for calculation of the proportion of days in the past year that were classified as “Extreme” weather. Because existing literature identifies extreme weather as the primary driver of flight cancellations in Nunavik, this proportion is used as a proxy for each airport’s expected cancellation rate.

3.2 Results Interpretation

Solving the model yields an optimal plan that balances ordering extra food (buffer) against the risk of spoilage along longer and more uncertain routes.

Key system totals:

- Total weekly demand: 107,392 lbs
- Total ordered volume: 109,158 lbs
- Safety stock (buffer) : 1,766 lbs
- System buffer ratio: 1.62%

In other words, the model recommends ordering a 1.6% buffer above theoretical weekly demand to guard against spoilage and delays. This enables the optimal solution to satisfy 100% of demand, while minimizing the need for costly buffer stock.

For the number of trucks the following result was obtained:

- La Grande (LGR) route:
 - Total cargo: 29,562 lbs
 - Total volume: 10.4 pallets
 - Trucks needed (continuous variable): 0.67 → planned as 1 truck per week
- Montreal (YUL) route:
 - Total cargo: 79,597 lbs
 - Total volume: 28.1 pallets
 - Trucks needed (continuous variable): 1.81 → planned as 2 trucks per week

The model is telling FCNQ that with the current network and demand, three trucks per week into the northern entry hub are enough to supply fresh fruits and vegetables to all 14 communities.

3.2.1 Buffer Stock Analysis

Rather than supplying a flat safety factor to every product, the model implicitly calculates a product specific buffer based on survival rate.

Across the entire network, the extra food ordered volume (buffer) is 1,766 lbs.

High spoilage items receive larger buffers:

- Bananas: 29,005 lbs demand → 30,101 lbs ordered → 3.78% buffer
- Grapes: 4,051 lbs demand → 4,146 lbs ordered → 2.34% buffer

Low spoilage items need little buffer:

- Turnips: 2836 lbs demand → 2849 lbs ordered → 0.46%
- Carrots: 16,204 lbs demand → 16,279 lbs ordered → 0.46%

The model is able to calculate the order size of each food items and avoids over-spending on extra food for resilient items like carrots and turnips.

3.2.2 Truck utilization and bottlenecks

The dual capacity constraints (weight & volume) diagnose what the trucks limiting factor is.

Using the optimal truck counts:

- LGR trucks:
 - Average weight per truck ~ 44,092 lbs
 - Average pallets per truck ~ 15.6 pallets
 - Weight utilization: ~ 100%
 - Volume utilization: ~ 60%
- YUL trucks:
 - Average weight per truck ~ 43,976 lbs
 - Average pallets per truck ~ 15.5 pallets
 - Weight utilization: ~ 99.7%
 - Volume utilization: ~ 60%

With the current mix of fresh produce, the trucks "Weigh Out" before they "Cube Out." The weight of dense items like carrots and apples hits the 44,092 lb limit while there is still floor space available. This suggests that future capacity planning for FCNQ must prioritize weight limits over volume when expanding the produce supply chain.

4. Problem Extensions

This section expands on the initial optimization model by incorporating deeper structural elements of FCNQ's operating environment. These proposed extensions draw directly from the model's limitations and assumptions, addressing gaps created by incomplete data, simplifying assumptions, and operational constraints. By refining these assumptions, the model can more accurately represent the complexity and uncertainty of northern supply chains.

Weather as the Only Disruptor

The model assumes weather is the sole cause of flight delays and cancellations, which simplifies disruption modeling. In practice, crews, aircraft availability, and operational issues also affect reliability. Excluding these additional factors understates true transit uncertainty.

Assumed Reliability of Truck Refrigeration

The model treats truck refrigeration as consistently reliable throughout transport. However, long driving distances, equipment variability, and pauses during driver rest breaks can interrupt cooling. This assumption may underestimate spoilage, especially for highly perishable food items.

Fixed Weekly Demand

Demand for each community is modeled as constant, despite natural week-to-week fluctuations.

Seasonal activities, hunting periods, or local events often change consumption needs. Assuming fixed demand limits the model's responsiveness to real shifts in community requirements.

Simplified Aircraft Loading Time

A single fixed 45-minute buffer is applied for aircraft loading at hubs. Actual loading times vary due to staffing levels, aircraft type, weather conditions, and congestion. This simplification reduces the model's ability to reflect realistic timing constraints.

Missing Cost and Operational Data

The model excludes financial costs and lacks detailed operational data such as refrigeration logs or mechanical disruption records. Without these inputs, economic trade-offs cannot be evaluated. This limits strategic planning and restricts the model to operational-level decisions only.

The following extensions directly address the limitations created by the model's current simplifying assumptions.

Extension 1: Time-Based Scheduling of Trucks and Flights

Incorporating a time-based structure would allow the model to coordinate truck departures with specific flight schedules rather than assuming a fixed loading buffer. This enhancement would capture variations in loading times, flight departure windows, and the risk of missed connections. By modelling transit hour-by-hour or in discrete time blocks, FCNQ could reduce idle time at airports, limit spoilage, and build more precise weekly operational plans.

Extension 2: Dynamic Parameters for Demand and Disruptions

Introducing dynamic, week-to-week parameters addresses the limitations of fixed demand and weather-only disruptions. Demand forecasts could vary seasonally or based on historical trends, while disruption probabilities could update as weather forecasts change. A dynamic or scenario-based model would more accurately reflect the uncertainty inherent in northern logistics, enabling FCNQ to adjust shipment quantities proactively and improve reliability during high-risk periods.

Extension 3: Enhanced Cost Logic and Strategic Evaluation

Integrating cost data transforms the model from a purely quantity-based optimization into a comprehensive decision-support tool. By including transportation costs, handling fees, spoilage

costs, and inventory carrying costs, FCNQ can evaluate the financial implications of different routing choices and shipping strategies. This expanded cost framework also enables analysis of long-term investments, such as improved refrigeration, added warehouse capacity, or dedicated transportation assets, to determine which options most effectively reduce both waste and total system cost.

5. Recommendations & Conclusions

The optimization model provides FCNQ with a structured and quantifiable method to determine how much perishable food should be shipped to remote northern communities each week. By incorporating spoilage rates, disruption probabilities, and capacity constraints, the model ensures communities receive the quantities they need while minimizing unnecessary over-ordering and reducing waste. This contributes directly to improving affordability and strengthening food security in a region where prices are already disproportionately high. To fully realize the benefits of this analytical approach, the following recommendations are proposed:

I. Integrate the Model Into FCNQ's Operational Workflow

The optimization model should be incorporated directly into FCNQ's weekly planning processes to ensure consistent use. Seamless integration will allow FCNQ to rely on the model's outputs when determining shipment quantities, truck allocations, and distribution strategies.

II. Develop a Centralized Application With Automated Data Updates

Asymmetric information is one of FCNQ's pain points. When interruptions occur and stakeholders have to decide what to do with the food, the food is still spoiling while the decisions are being made. A unified digital platform would enable stakeholders to simplify coordination, and improve the responsiveness of weekly planning.

III. Provide Detailed Operational Outputs for Decision-Makers

Model results should include not only the number of trucks required for each airport, but also recommended departure schedules and weekly logistics timelines. Enhanced visibility across the transportation network will help executives anticipate challenges and manage operations more effectively.

Northern food distribution is inherently complex, shaped by long distances, extreme weather, volatile flight networks, and high spoilage sensitivity. The optimization model developed in this report offers FCNQ a rigorous and data-driven method to determine weekly shipment quantities

while accounting for spoilage, capacity limits, and disruption probabilities. Extending the model to address its current assumptions such as fixed demand, and simplified timing, would significantly enhance its realism and decision-making power. When paired with robust operational data and integrated planning tools, the model can become a foundational element of FCNQ's strategy to improve food availability, reduce waste, and ultimately strengthen food security across all Nunavik communities.

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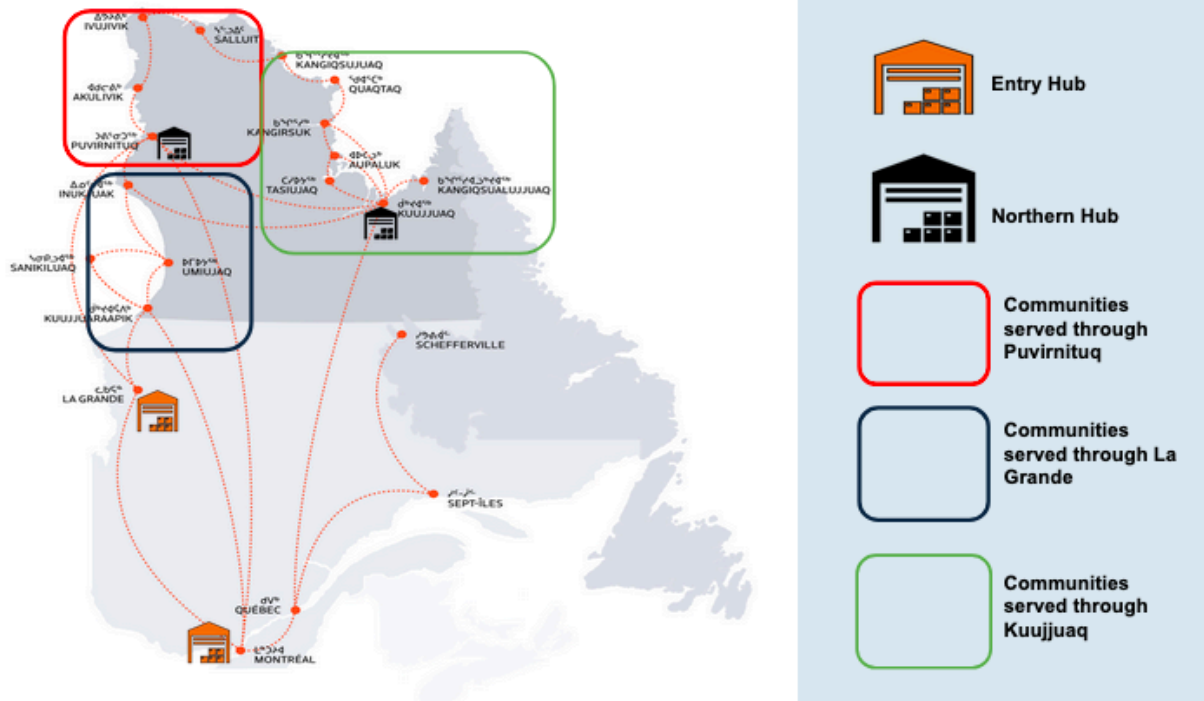
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Appendix

Appendix A



Appendix B

